

Element E: Application of STEM Principles and Practices

The proposed solution developed required extensive mathematical formulas in order to find the optimal speed of the flywheel. The process involved many experimental steps in order to figure out the correct speed and distance the frisbee launcher needed to produce. One way this was tested was by simply throwing a frisbee and then looking at the time the frisbee took to travel over a certain distance. The desired distance for the prototype to shoot the disc is 30 ft and using the time calculated, the speed of the flywheel came to 570 revolutions per minute. Using the calculated speed, we were able to find the optimal size for the flywheel and necessary axles.

Kinematics Methodology Used to Calculate Distance:

θ : *The angle the ramp makes with the horizontal surface*

$v_{initial}$: *Initial Velocity of Frisbee*

Horizontal Component Velocity: $v_{y0} = v_{initial} \cos(\theta)$

Determining the Horizontal Distance:

$$\Delta x = v_{y0}t + \frac{1}{2}at^2$$

The process used to produce the launcher supports STEM ideas as we incorporated the use of multiple manufacturing and programming techniques to ensure that effective designs could be produced. In addition, the prototype requires programming in order to effectively allow the user to change the speed of the motor. Computer science will be able to make the prototype to be more flexible to the user's needs as well as simplifying the use of the frisbee launcher for the students. Engineering was included throughout the entire process since the hardware portion of the frisbee launcher was constructed entirely through manufacturing processes and machines such as 3D printing and the use of CNC to produce complex designs and make the build easier for customization. Furthermore, the planning for the prototype involved many steps of the engineering process such as defining the problem and brainstorming solutions.

To ensure the validity of the proposed solution, an expert professor at George Mason University who specializes in engineering and scientific solutions reviewed our design. This expert gave valuable feedback as to what could be improved in the design of our frisbee launcher. He made sure to point out any possibilities of shortcomings with our design, given that the frisbee launcher was being designed for disabled students with limited motor skill capabilities. For example, he described how our frisbee launcher could have the possibility of being too big too heavy to operate for these children which provided great insight into the user-friendliness of the launcher. As a result, the feedback provided affected the design to be able to be placed fairly easily on top of a wheelchair or on a student's lap. The subject matter expert that was providing feedback is well qualified since instead of a stakeholder, he is an individual who can see both the manufacturing and user-friendliness side of the process. As a result, he was able to provide great feedback to improve the design and mechanics of the frisbee launcher.